1.Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

Objective: Configure AWS CloudTrail to:

- Capture all AWS API activities.
- Store CloudTrail logs in an Amazon S3 bucket.
- Send CloudTrail events to Amazon CloudWatch Logs.

Prerequisites

- AWS Account
- IAM User with AdministratorAccess
- AWS Region: us-east-1 (or your preferred region)

Step 1: Create an S3 Bucket

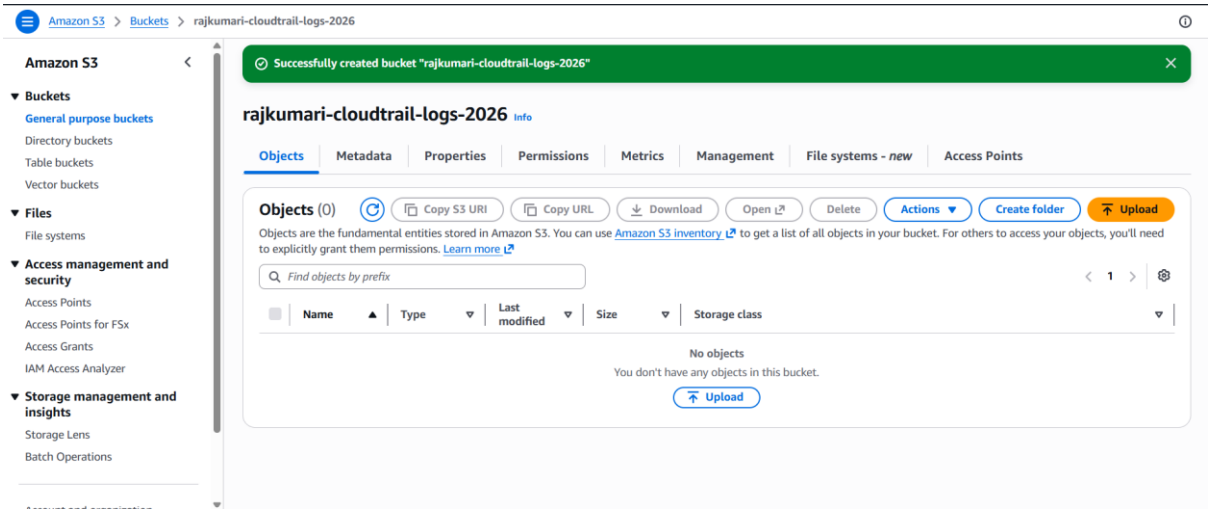
1. Login to AWS Console.
2. Search **S3**.
3. Click **Create bucket**.

Enter:

Setting	**Value**
Bucket Name	rajkumari-cloudtrail-logs-2026 (must be globally unique)
AWS Region	us-east-1
Block Public Access	Keep Enabled
Bucket Versioning	Optional

Setting	Value
Default Encryption	SSE-S3

Click **Create Bucket**.



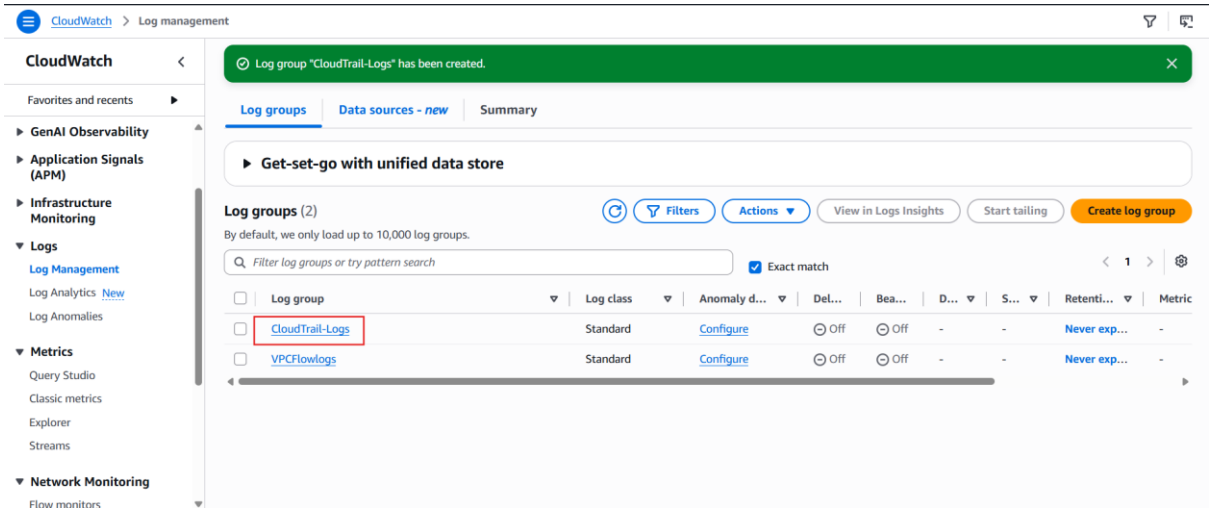
Step 2: Create a CloudWatch Log Group

1. Open **CloudWatch**.
2. Click **Logs**.
3. Select **Log Groups**.
4. Click **Create Log Group**.

Name: CloudTrail-Logs

Retention: Never Expire

Click **Create**.



Step 3: Create IAM Role for CloudTrail

Open: IAM

Click: Roles

Click: Create Role

Trusted Entity

Choose: AWS Service

Select: CloudTrail

Click: Next

Attach Permission

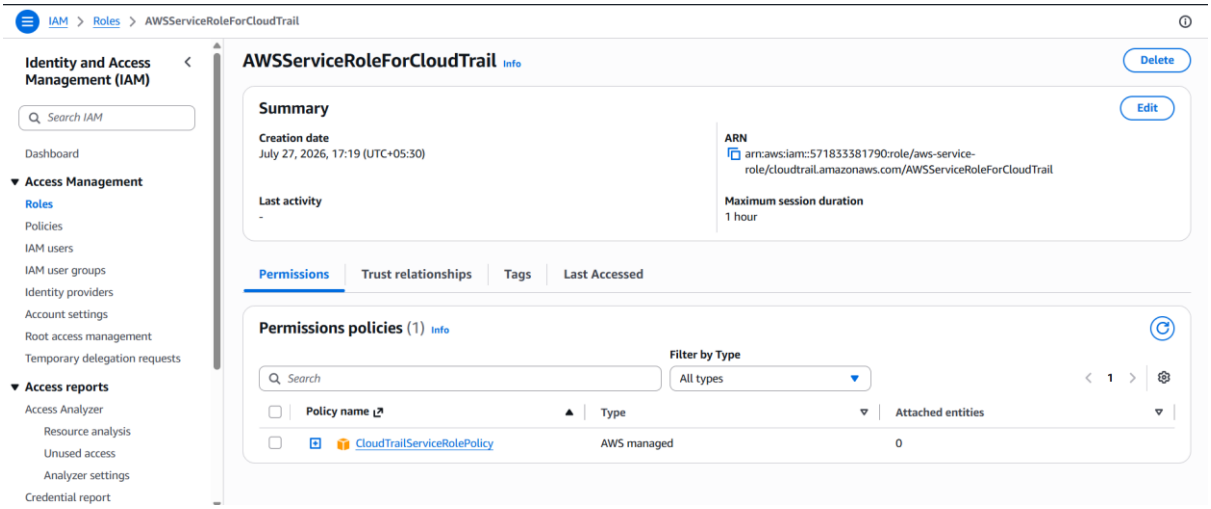
Search: CloudTrailServiceRolePolicy

Select it.

Click Next

Role Name: AWSServiceRoleForCloudTrail

Click: Create Role



Step 4: Open CloudTrail

Search

CloudTrail

Click

Trails

Click

Create Trail

Step 5: Configure CloudTrail

Trail Name

MyCloudTrail

Storage Location

Create new S3 bucket

or

Choose existing bucket

Select

rajkumari-cloudtrail-logs-2026

Leave

Log File SSE Encryption

Enabled.

Step 6: Configure Event Logging

Management Events

✓ Read

✓ Write

Data Events

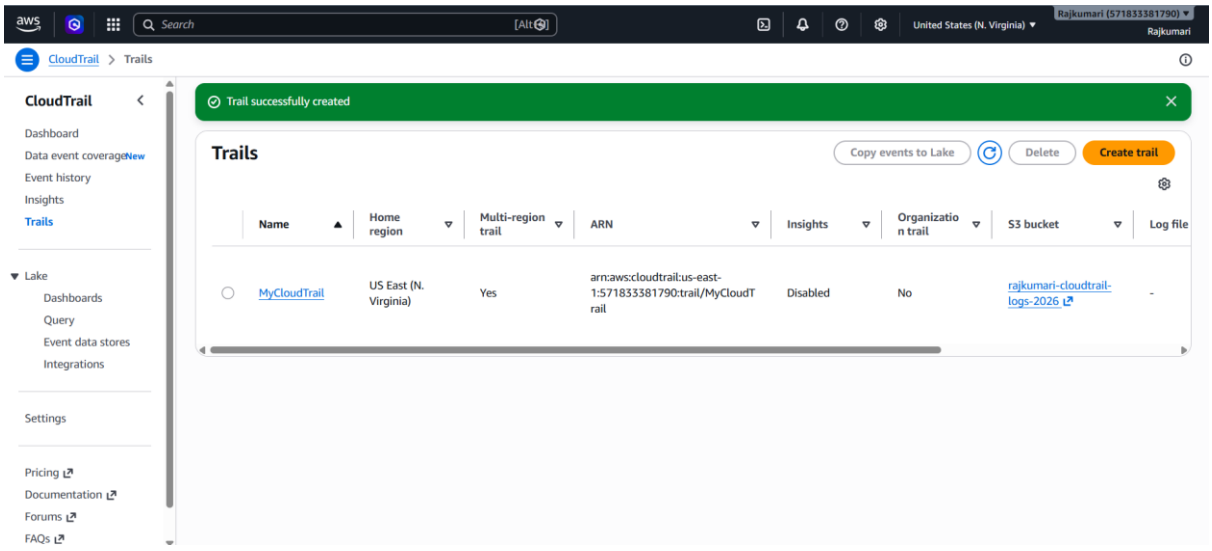
Skip

Insights

Skip

Click

Next



Step 7: Enable CloudWatch Logs

Enable

✓ CloudWatch Logs

Log Group

Choose

CloudTrail-Logs

IAM Role

Choose

CloudTrail-CloudWatch-Role

Click

Next

Step 8: Review

Review all settings.

Click

Create Trail

CloudTrail starts recording immediately.

Step 9: Verify Logs in S3

Go to

S3

Open

rajkumari-cloudtrail-logs-2026

Navigate to:

AWSLogs/

Account-ID/

CloudTrail/

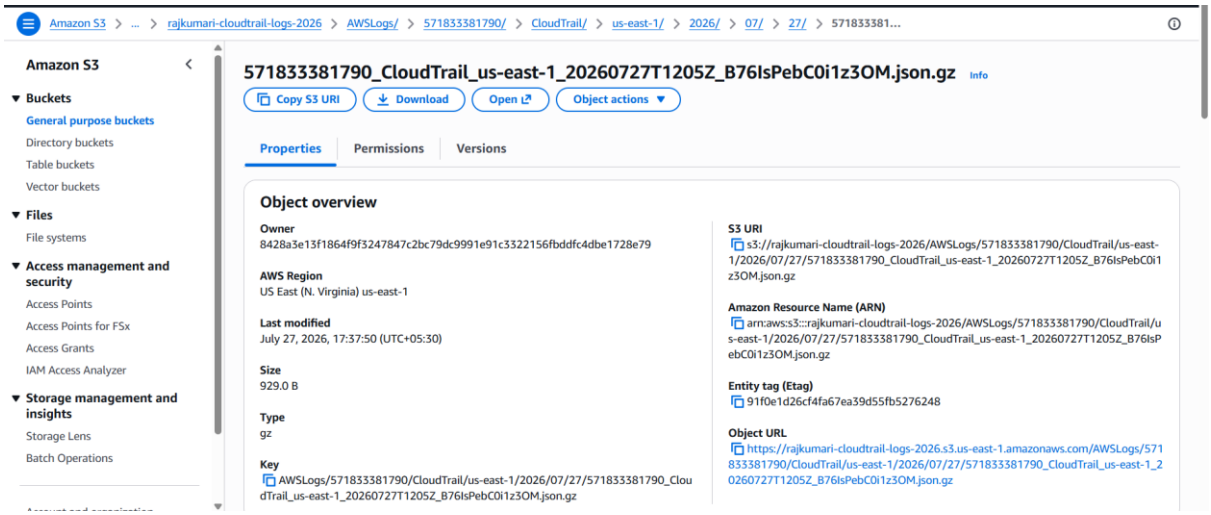
Region/

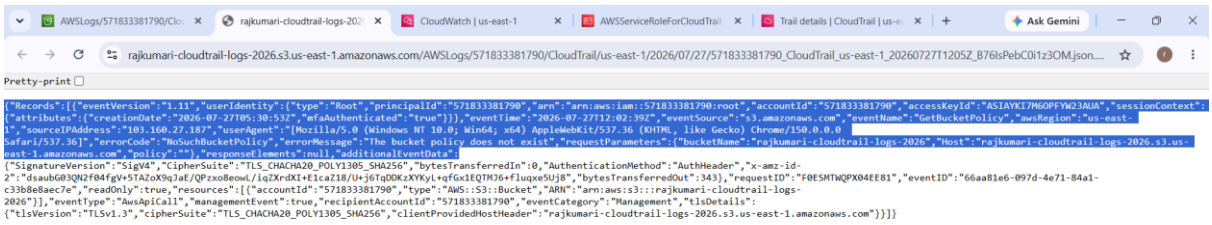
Year/

Month/

Day/

You should see compressed log files:





Step 10: Verify Logs in CloudWatch

Open

CloudWatch

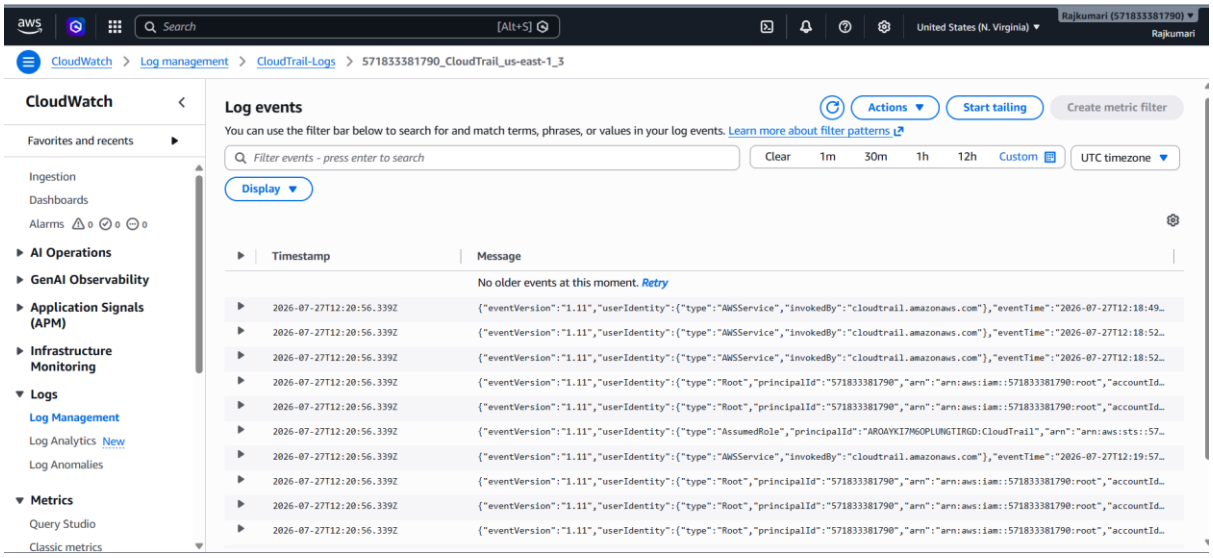
Go to

Logs

Open

CloudTrail-Logs

Open the latest **Log Stream**.



2. Enable SNS for CloudTrail to send alerts via email.

Objective: Configure **Amazon SNS (Simple Notification Service)** with **CloudTrail** so that whenever CloudTrail delivers a new log file, an email notification is sent to the subscribed email address.

Prerequisites

- AWS Account
- IAM User with AdministratorAccess
- Existing CloudTrail Trail
- Existing S3 Bucket for CloudTrail Logs
- Valid Email Address

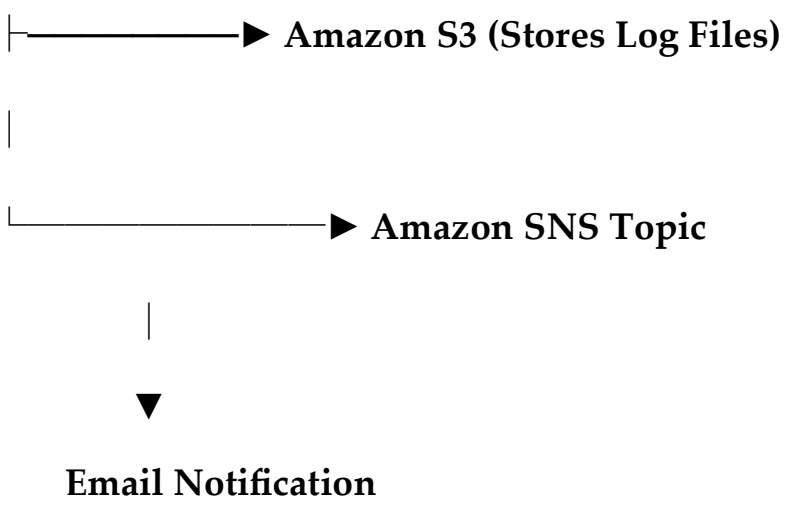
AWS API Activity

|

▼

AWS CloudTrail

|



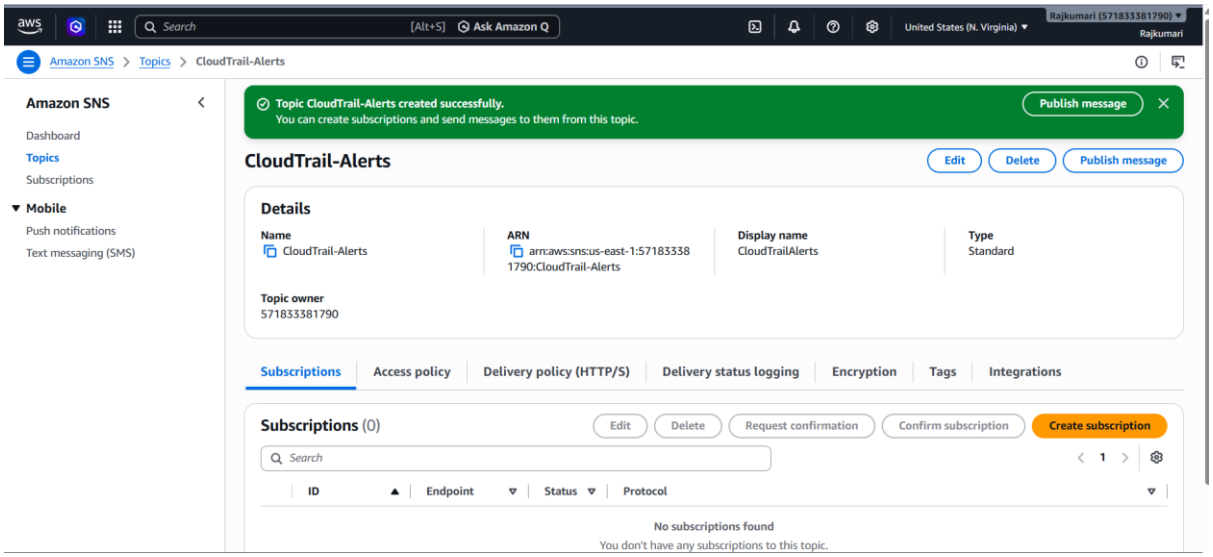
Step 1: Create an SNS Topic

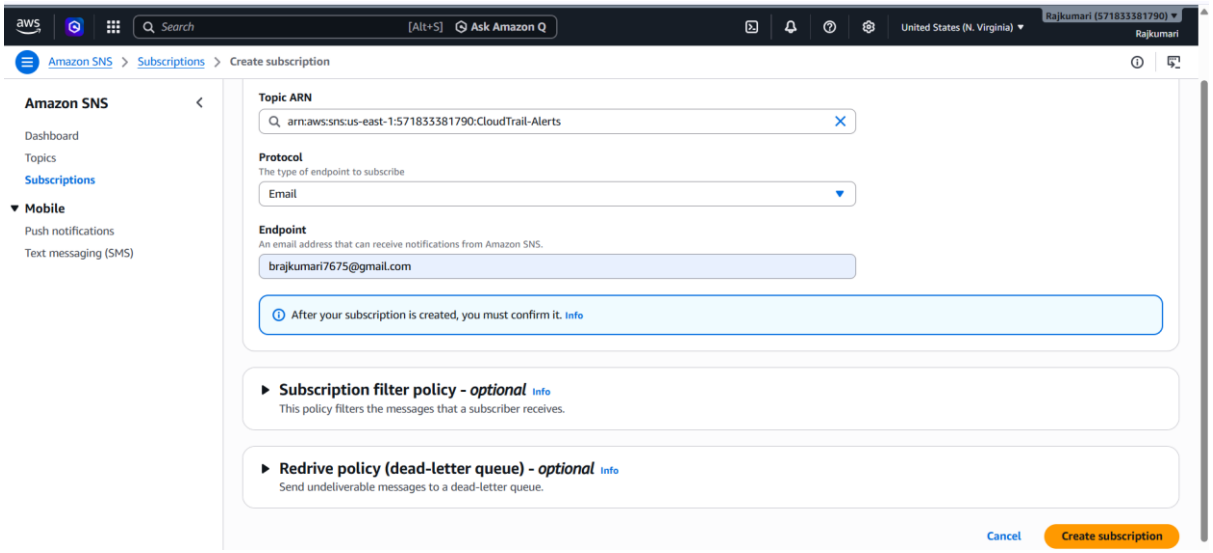
1. Sign in to the **AWS Management Console**.
2. Search for **SNS**.
3. Click **Simple Notification Service**.
4. In the left pane, click **Topics**.
5. Click **Create topic**.

Configure the following:

Setting	**Value**
Type	Standard
Name	CloudTrail-Alerts
Display Name (Optional)	CloudTrailAlerts

Click **Create topic**.

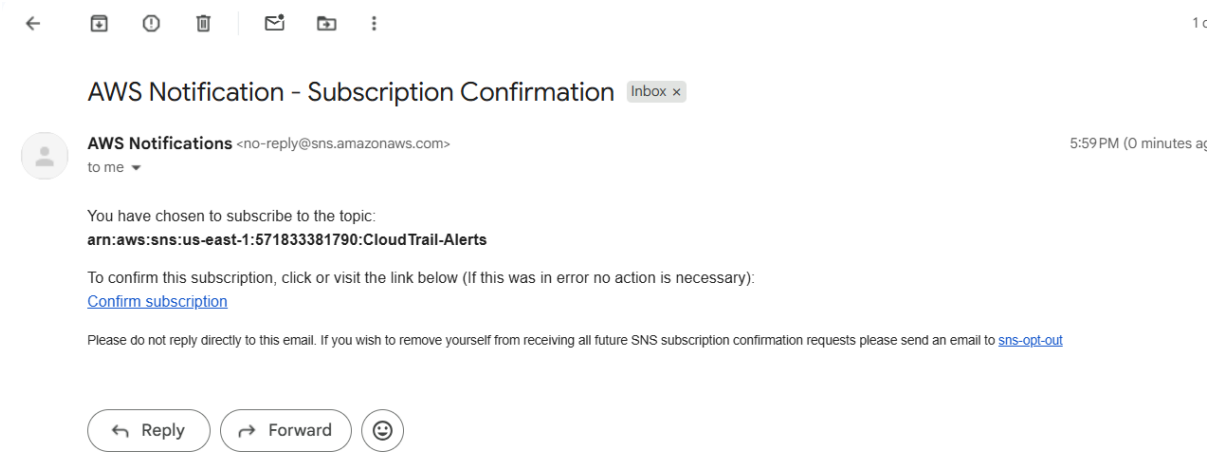




Step 3: Confirm the Subscription

1. Open your email inbox.
2. Look for an email from **AWS Notifications**.
3. Click **Confirm subscription**.

After confirmation, the subscription status changes from **Pending confirmation** to **Confirmed**.



Step 4: Configure CloudTrail

1. Open **CloudTrail**.
2. Click **Trails**.
3. Select your trail (**MyCloudTrail**).
4. Click **Edit**.

Step 5: Enable SNS Notifications

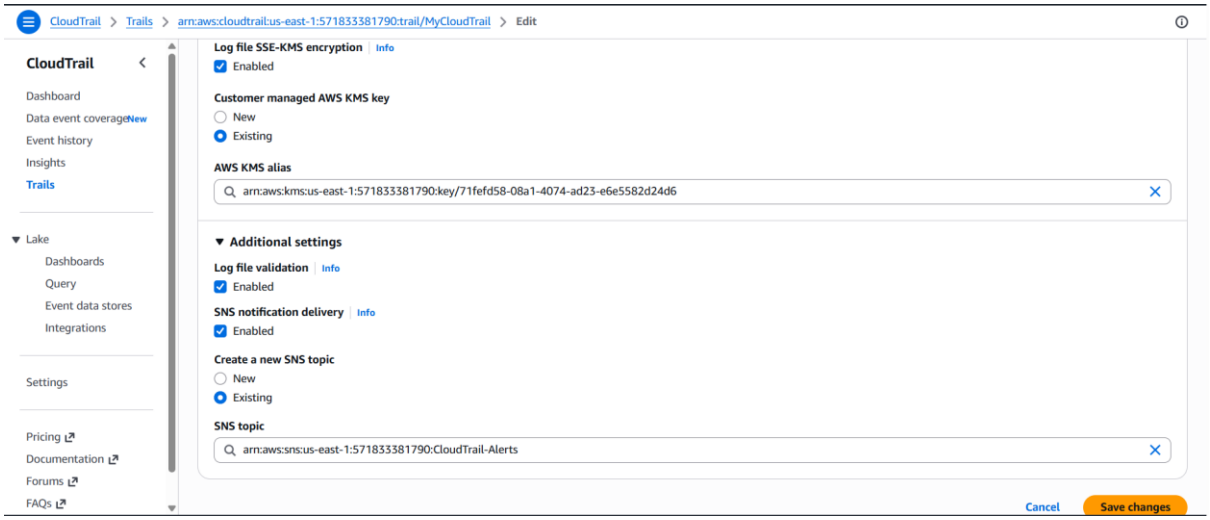
Scroll to the **SNS notification delivery** section.

Configure:

Setting	Value
SNS Notification	Enable

Setting	Value
SNS Topic	Existing
Topic Name	CloudTrail-Alerts

Click **Save changes**.



Step 6: Generate CloudTrail Activity

Perform some AWS actions, such as:

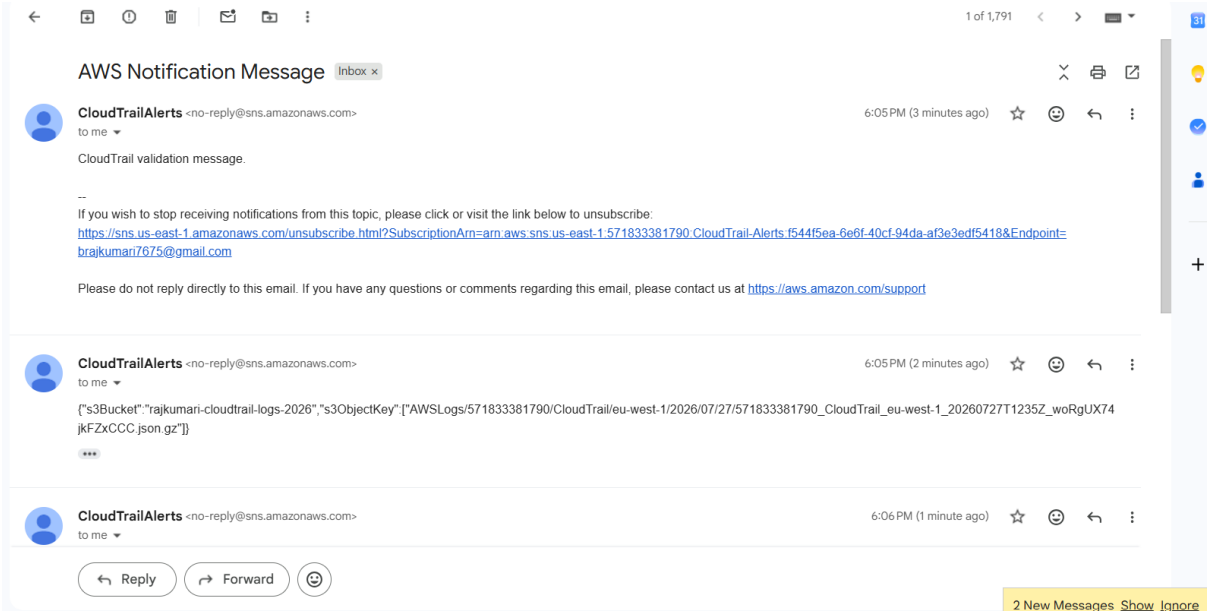
- Create an S3 bucket.
- Launch an EC2 instance.
- Create an IAM user.
- Stop or start an EC2 instance.

CloudTrail records these API calls.

Step 7: Verify Email Notification

Wait a few minutes.

Check your email inbox.



Step 8: Verify in the AWS Console

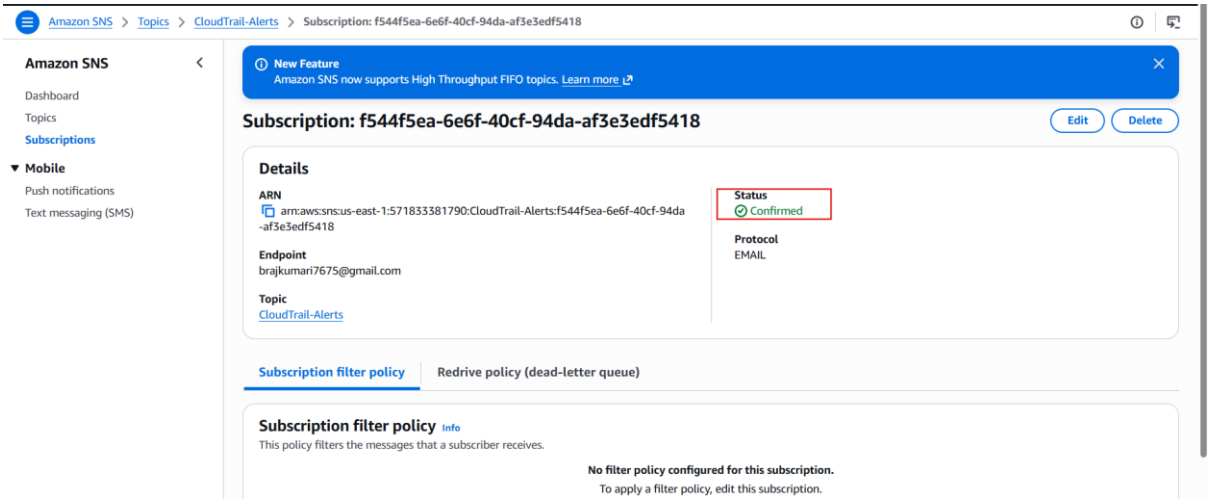
Verify SNS Subscription

Go to:

SNS → Topics → CloudTrail-Alerts → Subscriptions

Expected status:

Confirmed



Verify CloudTrail

Go to:

CloudTrail → MyCloudTrail

Expected:

SNS Notification Delivery

Enabled

AWS CLI Verification

3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

Objective: Configure **Amazon CloudWatch** to monitor an EC2 instance and record performance metrics such as:

- CPU Utilization
- Network In
- Network Out
- Disk Read Bytes
- Disk Write Bytes
- Status Check Failed

Prerequisites

- AWS Account

- IAM User with AdministratorAccess
- Running EC2 Instance
- CloudWatch Agent (Optional, for memory and disk metrics)

Step 1: Launch an EC2 Instance

If you already have an EC2 instance, skip this step.

1. Open **EC2 Console**.
2. Launch an Amazon Linux 2023 instance.
3. Instance Type: t2.micro (or t3.micro)
4. Create/select a key pair.
5. Configure the Security Group.
6. Launch the instance.

Step 2: Verify Basic Monitoring

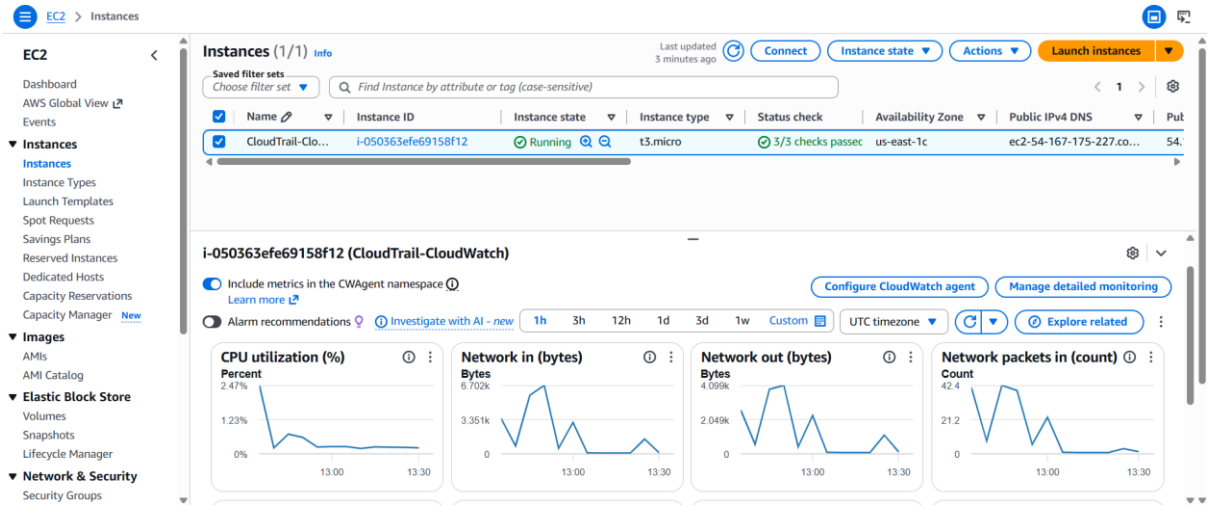
1. Open **EC2 Console**.
2. Select your EC2 instance.
3. Click the **Monitoring** tab.

You should see default CloudWatch metrics such as:

- CPU Utilization
- Network In
- Network Out
- Disk Read Bytes
- Disk Write Bytes

- Status Check Failed

Note: Basic monitoring collects metrics every **5 minutes**.



Step 3: Enable Detailed Monitoring (Optional)

1. Select the EC2 instance.
2. Click **Actions**.
3. Choose:

Monitor and troubleshoot

4. Click:

Manage detailed monitoring

5. Select:

Enable detailed monitoring

6. Save the changes.

Now CloudWatch collects metrics every **1 minute**.

Step 4: View Metrics in CloudWatch

1. Open **CloudWatch Console**.
2. Click **Metrics**.
3. Select:

All Metrics

4. Open:

EC2

5. Click:

Per-Instance Metrics

Select your instance.

Step 5: Monitor CPU Utilization

Locate the metric:

CPUUtilization

You can view:

- Average
- Minimum
- Maximum
- Sum

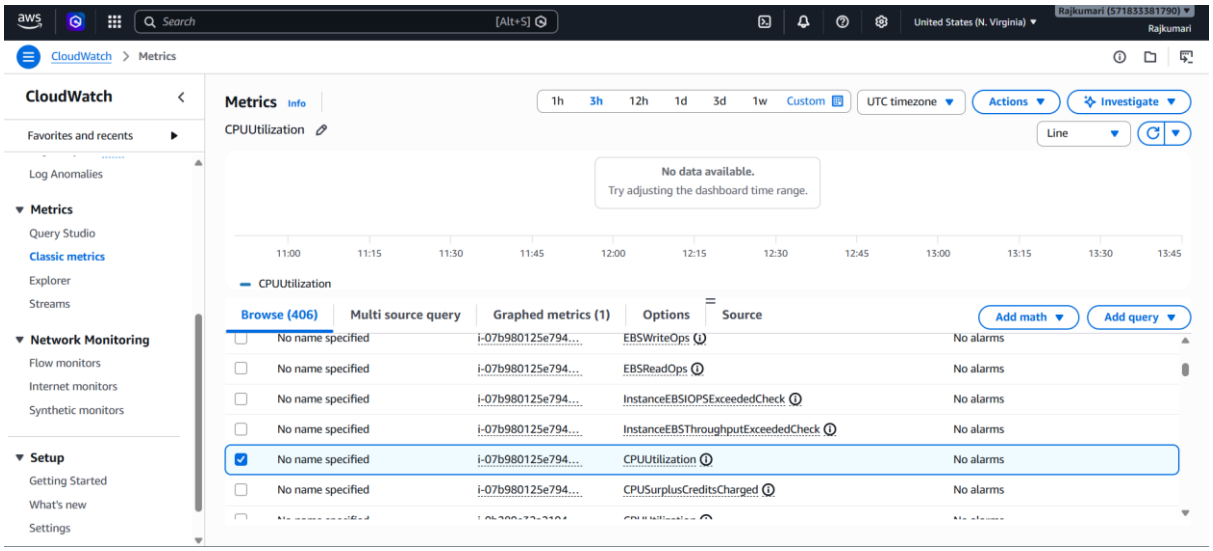
This graph shows the CPU usage percentage over tim

Step 6: Monitor Other Metrics

Observe the following metrics:

Metric	**Description**
CPUUtilization	CPU usage percentage
NetworkIn	Incoming network traffic
NetworkOut	Outgoing network traffic
DiskReadBytes	Bytes read from disk

Metric	Description
DiskWriteBytes	Bytes written to disk
StatusCheckFailed	EC2 instance health status



Step 7: Generate CPU Load

Connect to your EC2 instance:

ssh -i key.pem ec2-user@<Public-IP>

Install the stress tool:

Amazon Linux 2023

sudo dnf install stress -y

Run a CPU stress test for 5 minutes:

stress --cpu 2 --timeout 300

This increases CPU utilization.

```
[root@ip-172-31-18-188 ~]# sudo dnf install stress -y
Amazon Linux 2023 Kernel Livepatch repository   519 kB/s |  61 kB     00:00
Dependencies resolved.
================================================================================
 Package      Architecture Version                    Repository         Size
================================================================================
Installing:
 stress       x86_64       1.0.7-2.amzn2023.0.1       amazonlinux        34 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm          871 kB/s |  34 kB     00:00
--------------------------------------------------------------------------------
Total                                           425 kB/s |  34 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : stress-1.0.7-2.amzn2023.0.1.x86_64                     1/1
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64                     1/1
  Verifying        : stress-1.0.7-2.amzn2023.0.1.x86_64                     1/1

Installed:
  stress-1.0.7-2.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-18-188 ~]# stress --cpu 2 --timeout 300
stress: info: [28557] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
```

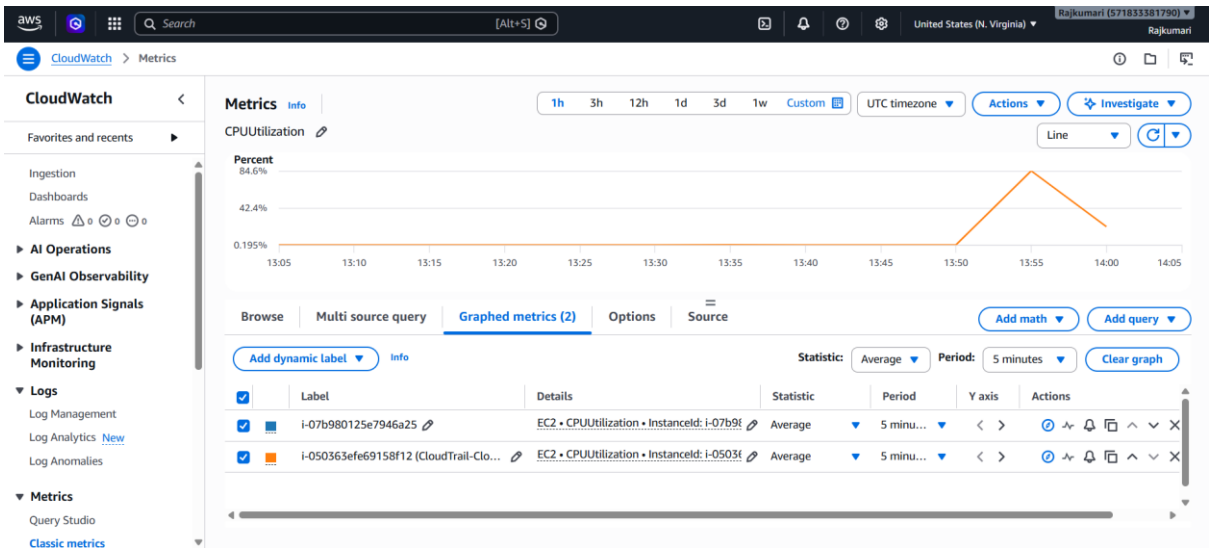
Step 8: Verify CPU Graph

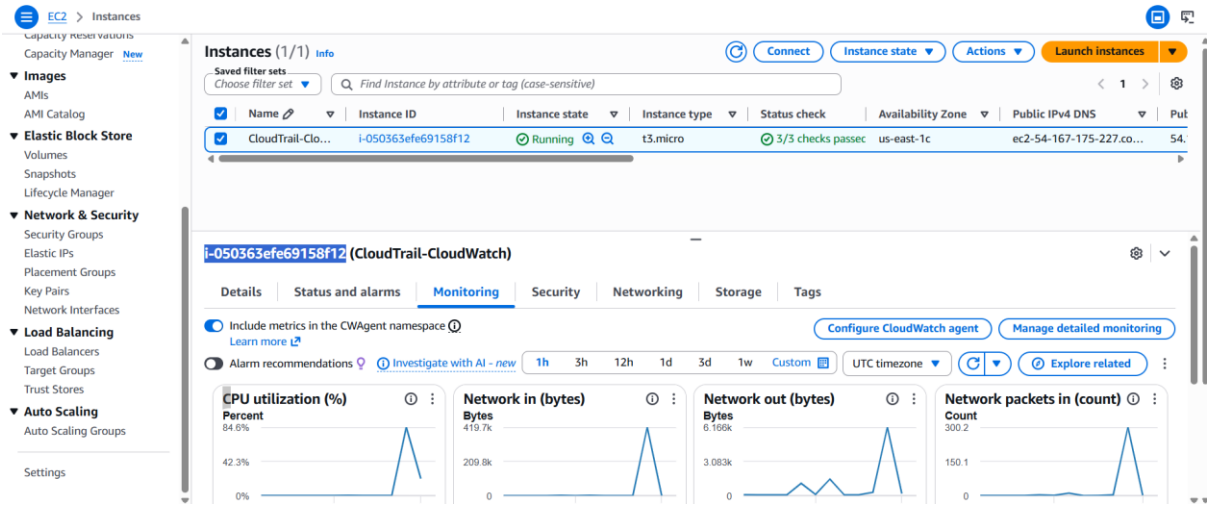
Go back to:

CloudWatch → Metrics → EC2 → Per-Instance Metrics

Refresh the graph after a few minutes.

You should observe an increase in **CPUUtilization**.





4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Objective: Create a CloudWatch alarm that sends an email notification through Amazon SNS when the EC2 instance's CPU utilization is greater than **70%**.

Step 1: Open CloudWatch

1. Sign in to the AWS Management Console.
2. Search for **CloudWatch**.
3. Open the **CloudWatch** service.

Step 2: Go to Alarms

In the left navigation pane:

CloudWatch → Alarms → All alarms

Click: **Create alarm**

Step 3: Select Metric

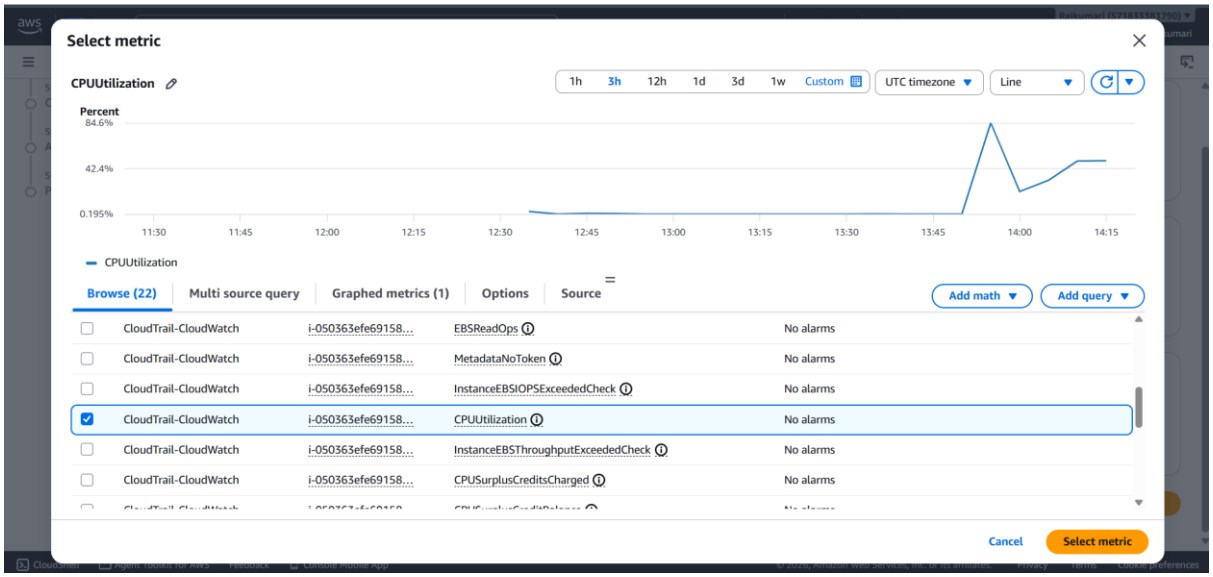
1. Click **Select metric**.
2. Choose:

EC2

3. Click:

Per-Instance Metrics

4. Select your EC2 instance's **CPUUtilization** metric.
5. Click **Select metric**.



Step 4: Configure the Metric

Configure the alarm as follows:

Setting	Value
Metric	CPUUtilization
Statistic	Average
Period	5 Minutes

Click **Next**.

Step 5: Configure the Threshold

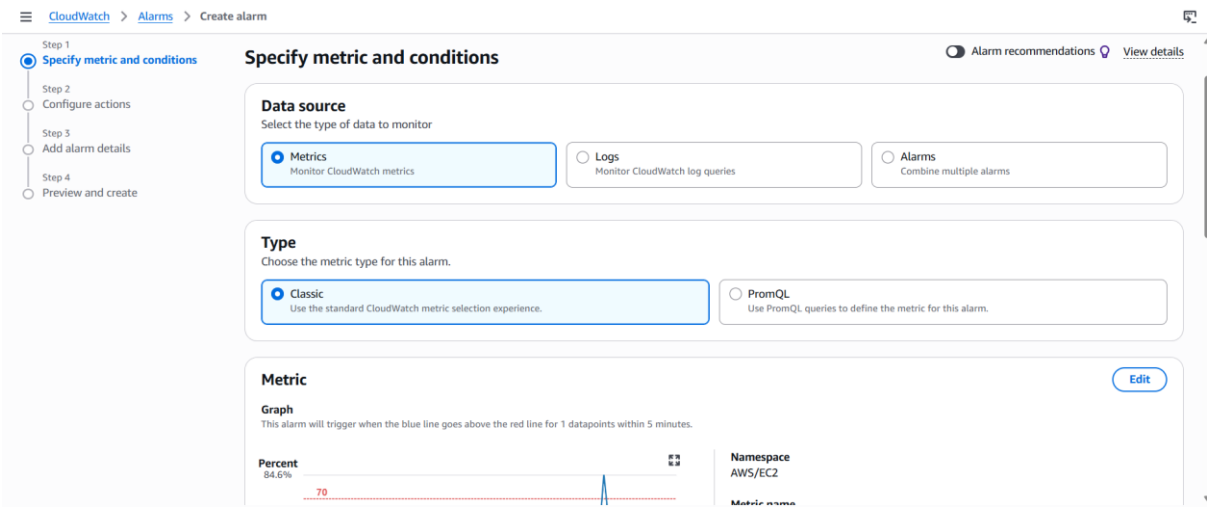
Set the following values:

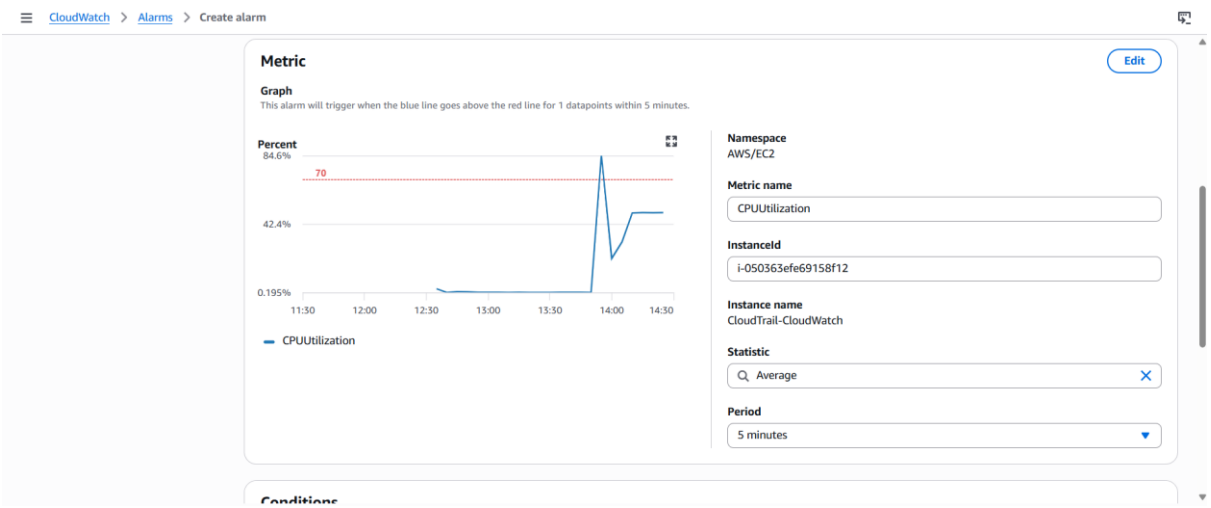
Setting	Value
Threshold Type	Static
Condition	Greater
Threshold Value	**70**

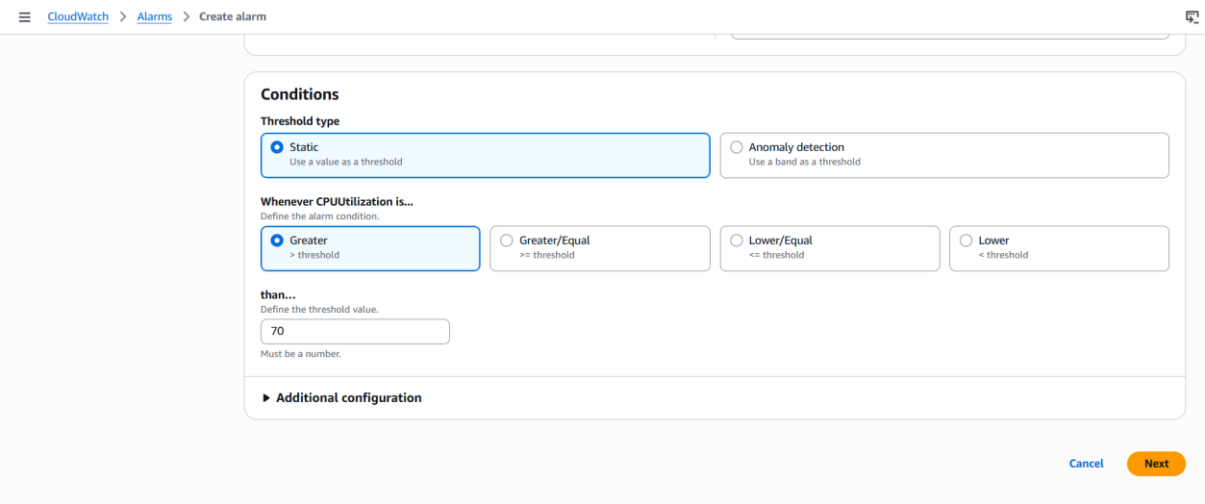
This means:

Trigger the alarm if the average CPU utilization is **greater than 70%**.

Click **Next**.







Step 6: Configure Notifications

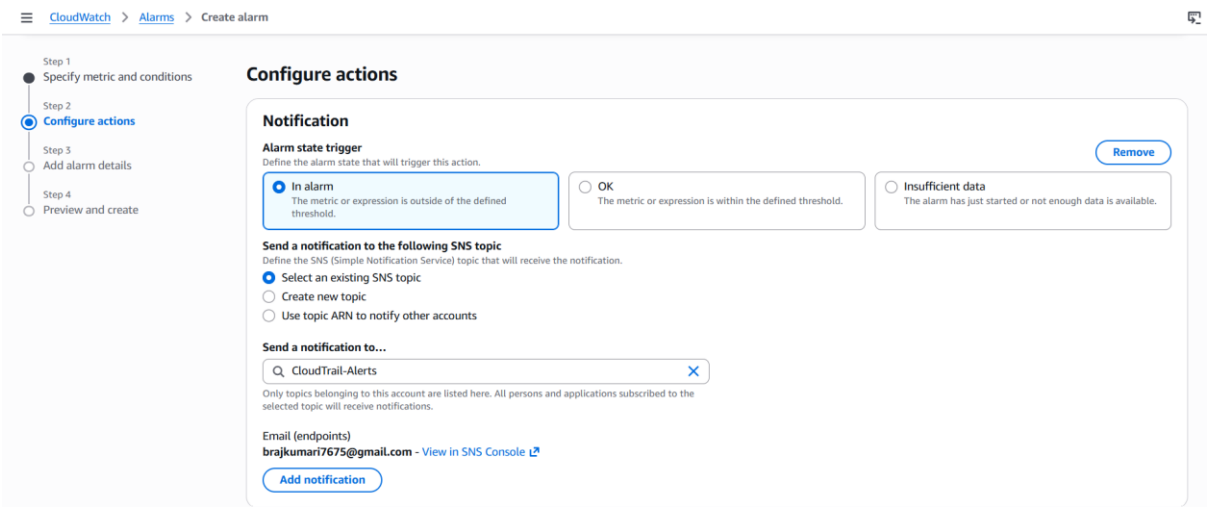
Under **Notification**:

- **Alarm state trigger:** In alarm
- **Send a notification to the following SNS topic:**
    - Choose **Select an existing SNS topic**
    - Select your SNS topic (for example, CloudTrail-Alerts, or create a new topic such as EC2-CPU-Alerts if you prefer to keep it separate)

If you don't already have an email subscription:

1. Create a new SNS topic.
2. Add your email address.
3. Confirm the subscription from your email inbox.

Click **Next**.

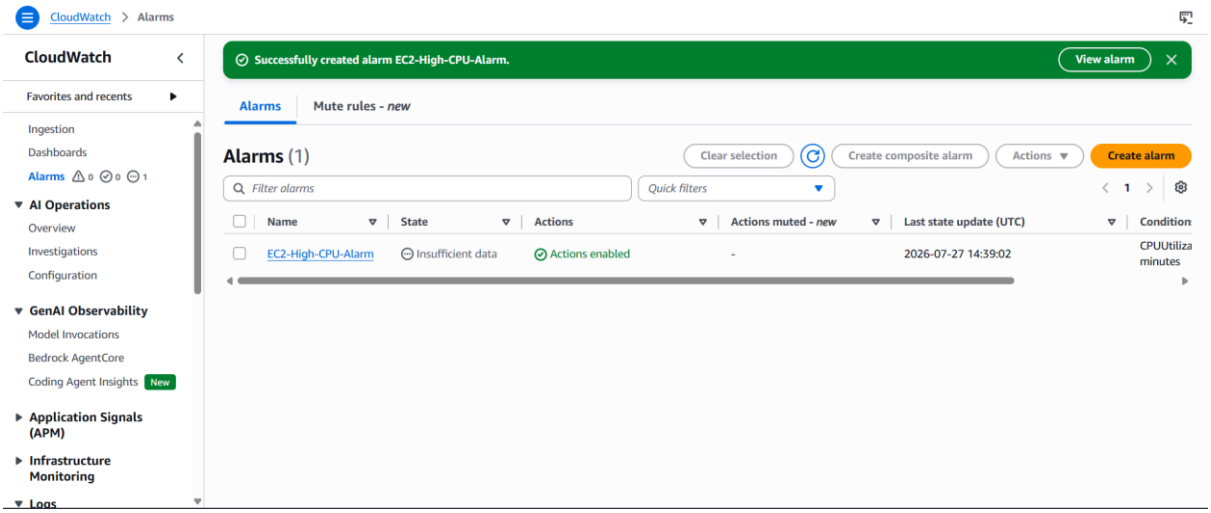


Step 8: Review and Create

Review all the settings.

Click:

Create alarm



Step 9: Verify the Alarm

Go to:

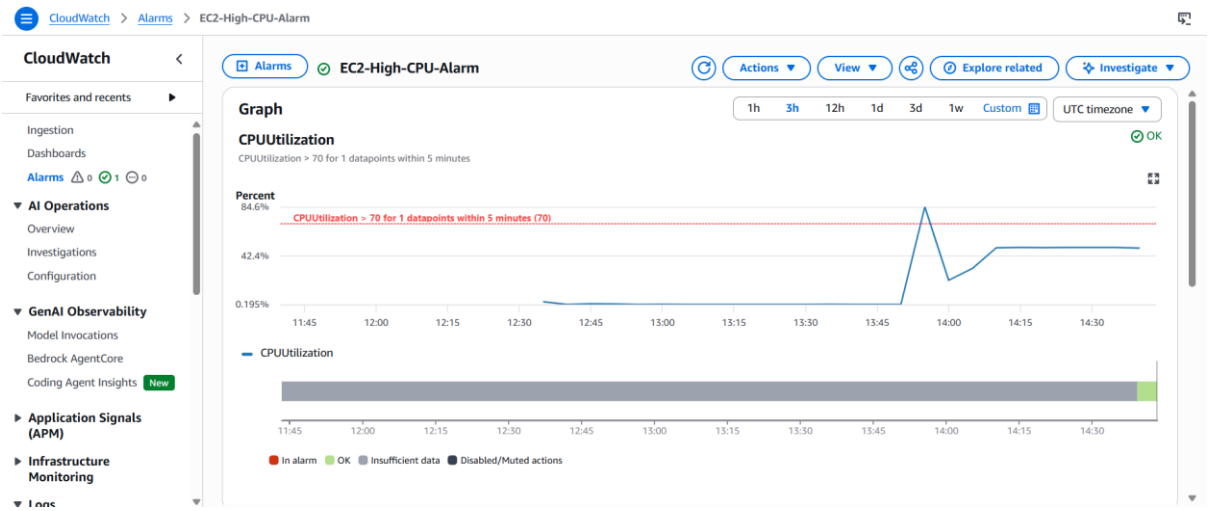
CloudWatch → Alarms

Initially, the alarm state will be: ok

After CPU utilization stays above **70%** for the configured evaluation period, the state changes to: ALARM

You should also receive an email notification from Amazon SNS.

When the CPU utilization drops below the threshold, the alarm automatically returns to the **OK** state.



5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert

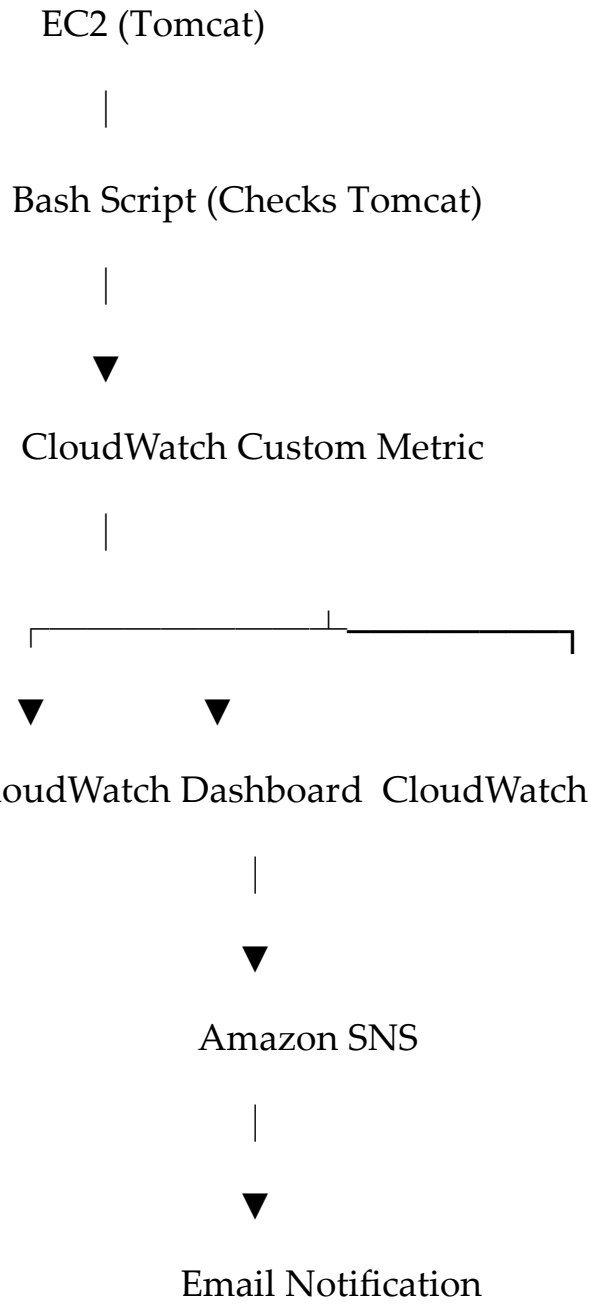
Objective

- Create a CloudWatch Dashboard.
- Monitor whether the Tomcat service is **running**.
- Send an email alert if the Tomcat service **stops**.

Prerequisites

- Running EC2 instance
- Apache Tomcat installed
- Amazon CloudWatch Agent installed
- IAM Role attached to EC2 with:
    - CloudWatchAgentServerPolicy
    - AmazonSSMManagedInstanceCore (recommended)
- SNS Topic with confirmed email subscription

Architecture



Step 1: Verify Tomcat

Connect to the EC2 instance: $ sudo systemctl status tomcat

If Tomcat is running, you should see: Active: active (running)

```
[root@ip-172-31-18-188 opt]# cd /opt/tomcat/bin
[root@ip-172-31-18-188 bin]# chmod +x *.sh
[root@ip-172-31-18-188 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli
.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-18-188 bin]#
```

Step 2: Create Tomcat Monitoring Script

Create a script that checks Tomcat status.

sudo vi /home/ec2-user/tomcat_status.sh

```
#!/bin/bash

STATUS=$(systemctl is-active tomcat)

if [ "$STATUS" = "active" ]
then
echo "1"
else
echo "0"
fi
~
```

Step 3: Install CloudWatch Agent

Install agent: $ sudo yum install amazon-cloudwatch-agent -y

Check: sudo systemctl status amazon-cloudwatch-agent

```
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
     Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-07-27 15:14:09 UTC; 29s ago
   Main PID: 31607 (amazon-cloudwat)
      Tasks: 7 (limit: 1059)
     Memory: 46.3M
        CPU: 602ms
     CGroup: /system.slice/amazon-cloudwatch-agent.service
             └─31607 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazo

Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31613]: Executing /opt/aws/amazon-cloudwatch-agent/bin/a
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: D! [EC2] Found active network interface
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: I! imds retry client will retry 1 timesI! Detect
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: 2026/07/27 15:14:10 Reading json config file pat
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: /opt/aws/amazon-cloudwatch-agent/etc/amazon-clou
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: 2026/07/27 15:14:10 Reading json config file pat
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: 2026/07/27 15:14:10 I! Valid Json input schema.
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: I! Trying to detect region from ec2
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31618]: 2026/07/27 15:14:10 Configuration validation fir
Jul 27 15:14:10 ip-172-31-18-188.ec2.internal start-amazon-cloudwatch-agent[31607]: I! Detecting run_as_user...
```

6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Step 1: Install Nginx

Connect to your EC2 instance:

Install Nginx: $ sudo yum install nginx -y

Step 2: Start Nginx Service

Start Nginx: $ sudo systemctl start nginx

Check status: $sudo systemctl status nginx

Step 3: Create Nginx Monitoring Script

Create a script: $ sudo vi /home/ec2-user/nginx_status.sh

Step 4: Configure CloudWatch Agent

Create CloudWatch Agent configuration:

root@ip-172-31-18-188:/opt/tomcat/bin

```
[root@ip-172-31-18-188 bin]# sudo vi /home/ec2-user/nginx_status.sh
[root@ip-172-31-18-188 bin]# chmod +x /home/ec2-user/nginx_status.sh
[root@ip-172-31-18-188 bin]# sudo vi /opt/aws/amazon-cloudwatch-agent/etc/amazon
-cloudwatch-agent.json
[root@ip-172-31-18-188 bin]# sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cl
oudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json \
-s
****** processing amazon-cloudwatch-agent ******
Starting config-downloader, this will map back to a call to amazon-cloudwatch-ag
ent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with argu
ments: [config-downloader -download-source file:/opt/aws/amazon-cloudwatch-agent
/etc/amazon-cloudwatch-agent.json -output-dir /opt/aws/amazon-cloudwatch-agent/e
tc/amazon-cloudwatch-agent.d -config /opt/aws/amazon-cloudwatch-agent/etc/common
-config.toml -multi-config default -mode ec2]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
Start configuration validation...
2026/07/27 15:26:40 Reading json config file path: /opt/aws/amazon-cloudwatch-ag
ent/etc/amazon-cloudwatch-agent.d/file_amazon-cloudwatch-agent.json.tmp ...
2026/07/27 15:26:40 I! Valid Json input schema.
2026/07/27 15:26:40 Configuration validation first phase succeeded
Starting config-translator, this will map back to a call to amazon-cloudwatch-ag
ent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with argu
ments: [config-translator -config /opt/aws/amazon-cloudwatch-agent/etc/common-co
nfig.toml -multi-config default -input /opt/aws/amazon-cloudwatch-agent/etc/amaz
on-cloudwatch-agent.json -input-dir /opt/aws/amazon-cloudwatch-agent/etc/amazon-
cloudwatch-agent.d -output /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatc
h-agent.toml -mode ec2]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config
 /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
Configuration validation second phase succeeded
Configuration validation succeeded
[root@ip-172-31-18-188 bin]#
```

CloudTrailAlerts <no-reply@sns.amazonaws.com> 8:55 PM (2 minutes ago)
to me

{"s3Bucket":"rajkumari-cloudtrail-logs-2026","s3ObjectKey":["AWSLogs/571833381790/CloudTrail/us-east-1/2026/07/27/571833381790_CloudTrail_us-east-1_20260727T1525Z_gpxvWmIpLMRYwl0D.json.gz"]}

CloudTrailAlerts <no-reply@sns.amazonaws.com> 8:57 PM (1 minute ago)
to me

{"s3Bucket":"rajkumari-cloudtrail-logs-2026","s3ObjectKey":["AWSLogs/571833381790/CloudTrail/us-east-1/2026/07/27/571833381790_CloudTrail_us-east-1_20260727T1530Z_EMPDxTQqkNzc7H9p.json.gz"]}

CloudTrailAlerts <no-reply@sns.amazonaws.com> 8:57 PM (1 minute ago)
to me

{"s3Bucket":"rajkumari-cloudtrail-logs-2026","s3ObjectKey":["AWSLogs/571833381790/CloudTrail/us-east-1/2026/07/27/571833381790_CloudTrail_us-east-1_20260727T1525Z_Q9clFonZiYuiKhYE.json.gz"]}

Reply Forward